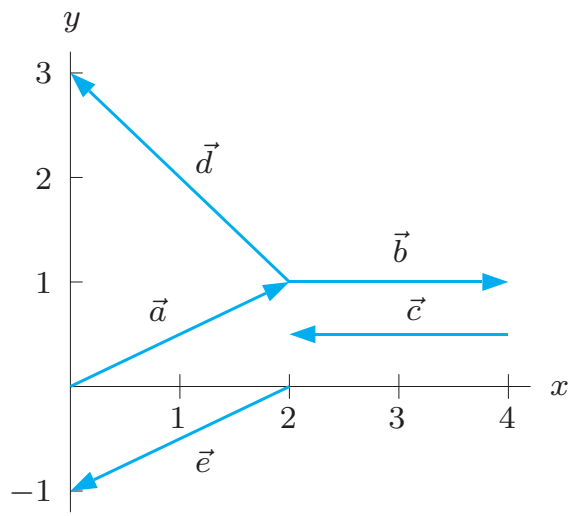
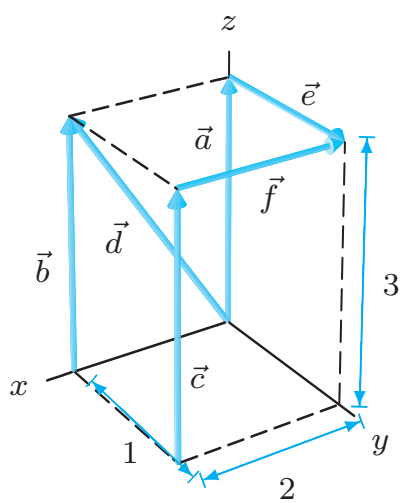
Suggested problems: 1-33 odd

Problem 1: Resolve the following vectors into the basis vectors.



Problem 2: Resolve the following vectors into the basis vectors.



Problem 3: Perform the following operations.

(a) $-4(\vec{i} - 2\vec{j}) + (-3)(2\vec{i} + \vec{j})$

(b) $\frac{1}{2}(2\vec{i} - \vec{j} + 3\vec{k}) + 3(\vec{i} - \frac{1}{6}\vec{j} + \frac{1}{2}\vec{k})$

Problem 4: Find a vector with length 2 which points in the same direction as $\vec{v} = \vec{i} - \vec{j} + 2\vec{k}$.

Problem 5: Find the values of a making $\vec{v} = 5a\vec{i} - 3\vec{j}$ parallel to $\vec{w} = a^2\vec{i} + 6\vec{j}$.